

Xinxuan Liu

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EDUCATION

Ph.D. in Economics, University of Illinois Urbana-Champaign *Expected May 2027*
Committee: Eun Yi Chung (Co-Chair), Ji Hyung Lee (Co-Chair), Joshua Shea, Drew Creal
M.S. in Policy Economics, University of Illinois Urbana-Champaign 2022
B.Econ. in Financial Engineering, Hefei University of Technology 2020

RESEARCH FIELDS

Primary Field: Econometric Theory (Treatment Effects, Nonparametric Inference)
Secondary Fields: Time Series Forecasting, Machine Learning

JOB MARKET PAPER

Beyond the Average and the Observed: Distributional Tests of Treatment Effect Heterogeneity in Regression Discontinuity Designs

Abstract: This paper proposes distributional tests of treatment effect heterogeneity in regression discontinuity designs. While existing tests compare average effects across subgroups defined by observed covariates, our tests instead detect heterogeneity anywhere in the outcome distribution and require no choice of covariates, so they also capture heterogeneity driven by unobservables. The tests ask whether the treated outcome distribution at the cutoff is a constant shift of the control distribution. We compare the two distributions after aligning them by the estimated shift and obtain critical values by permutation inference. When the shift is known, the permutation test is exact in finite samples; when it must be estimated, a Durbin problem arises and the test is no longer valid. We provide two remedies: a Khmaladze martingale transformation that removes the drift induced by estimation, and a prepivoted permutation test that reproduces that drift in the reference distribution. Both are asymptotically valid. In Monte Carlo simulations, the two tests hold their nominal size not only in large samples but also in small ones, and they have power against latent heterogeneity. We apply the tests to the effect of Islamic municipal rule on female education in Turkey and find evidence of heterogeneity within more religious municipalities that a comparison of average effects would miss.

WORKING PAPERS

Learning from Déjà Vu: Contextual Bandit-Based Model Selection for Macroeconomic Nowcasting

Selects among nowcasting models by finding the past periods whose financial and news conditions most resemble the present and choosing the model that nowcast them best. In real-time U.S. GDP nowcasting from 2006 to 2025, it improves on every benchmark feasible in real time, with the gains concentrated in the pandemic quarters.

Nowcasting Inflation with a Covariate Foundation Model

Nowcasts inflation with a pretrained transformer that reads high-frequency energy prices as exogenous inputs. It outperforms the classical mixed-frequency benchmarks on both point and density metrics, and with the month only partly observed it already surpasses every benchmark's full-month accuracy.

PRESENTATIONS

Job Market Seminar, University of Illinois Urbana-Champaign 2026
Econometrics Lunch Seminar, University of Illinois Urbana-Champaign 2024

RESEARCH EXPERIENCE

- Research Assistant**, University of Illinois Urbana-Champaign (Prof. Eun Yi Chung) 2023
Developed R code for Monte Carlo studies of quantile inference for time series, evaluating studentized and self-normalized statistics for permutation tests.
- Research Assistant**, University of Illinois Urbana-Champaign (Prof. Mark Borgschulte) 2022–2023
Built a gender-norms index from General Social Survey data, matched spouses via household rosters, and produced regression kink figures and descriptive analyses for an applied microeconomics project.

TEACHING EXPERIENCE

- Teaching Assistant**, University of Illinois Urbana-Champaign 2022–2026
Designed and taught discussion sections, wrote practice problems, and held office hours.
 - Economic Statistics I: Spring 2023, Fall 2024, Fall 2025, Spring 2026
 - Microeconomic Principles: Fall 2022*Facilitating Learning Excellence (FLEX) Award (Spring 2026)*
List of Teachers Ranked as Excellent by Their Students (Spring 2023, Fall 2024)
- Course Assistant**, University of Illinois Urbana-Champaign 2023–Present
Prepared exam solutions, ran review sessions, and graded coursework.
 - Intermediate Econometrics: Fall 2026
 - Econometrics of Policy Evaluation: Spring 2025
 - Time Series Analysis (graduate): Fall 2024
 - Financial Economics (graduate): Fall 2024
 - Econometric Analysis I (Ph.D.): Fall 2023
 - Financial Econometrics: Fall 2023

PROFESSIONAL SERVICE

- Reviewer and Organizing Assistant, Midwest Econometrics Group Conference 2025

SKILLS

Programming: R, Python, MATLAB, Stata, SQL, LaTeX; experience with HPC clusters and AWS
Languages: English (fluent), Mandarin Chinese (native)

REFERENCES

Eun Yi Chung (Co-Chair)
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